Notes: Griffin & Tang (JF, 2012)
Did Subjectivity Play a Role in CDO Credit Ratings?

Contents

1 Big picture

- **Question.** Did subjective adjustments by credit rating agencies inflate CDO credit ratings before the financial crisis?
- **Object studied.** The paper studies 916 CDOs issued from January 1997 to December 2007, with detailed information from a leading CRA's model outputs and surveillance reports.
- **Main variable.** The key variable is the **AAA adjustment**: the fraction of a CDO actually rated AAA minus the fraction implied by the CRA's own main quantitative model.
- **Core fact.** The average CDO has 75.5% of its capital structure rated AAA, while the CRA model implies only 63.4%. The average adjustment is therefore 12.1 percentage points.
- **Direction of adjustment.** 84.6% of adjustments are positive. The average adjustment rises over time, reaching 18.2% in 2007.
- **Pattern.** Adjustments are largest when the model-implied AAA fraction is smallest. This compresses very different CDOs into similar actual AAA tranche sizes.
- **Determinants.** Standard CDO features, manager experience, insurance, liquidity support, overcollateralization, alternate models, and excess spread do not explain much of the adjustments.
- **Consequences.** Larger AAA adjustments predict more severe subsequent downgrades, including downgrades to CC or below.
- **Second rating issue.** Before April 2007, actual default-rate criteria used for AAA CDO ratings were often looser than the CRA's publicized AAA default-rate criteria.
- **Magnitude.** The paper estimates economically large valuation effects. Depending on method, the value impact is about \$14.7 million, \$42.2 million, or \$94.1 million per CDO.
- **Takeaway.** The paper provides direct evidence that rating outcomes were not simply mechanical outputs of the rating model; subjective adjustments and criterion deviations materially increased AAA tranche sizes.

2 Data and measurement

2.1 CDO sample

- The sample contains 916 CDOs issued from January 1997 to December 2007.
- The total note face value is about $612.8 billion.
- The CDO categories are CBOs, CLOs, ABS CDOs, and CDO^2.
- Data come from the first CRA surveillance reports, new-issue reports, presale reports, and rating databases.
- The important feature of the data is that they contain the CRA model output, not only the final credit rating.

Interpretation: The data let the authors compare final ratings against what the CRA's main quantitative model itself would have assigned. This is much stronger than inferring rating inflation only from ex post defaults.

2.2 CDO collateral and liability variables

The main CDO-level variables are:

- collateral default probability;
- collateral average rating;
- collateral average maturity;
- collateral size;
- number of assets and number of obligors;
- synthetic CDO indicator;
- collateral-manager deal experience;
- overcollateralization ratio;
- insurance and liquidity-support indicators;
- actual AAA fraction.

Interpretation: These controls proxy for the variables that could rationally justify a larger or smaller AAA tranche. If the adjustments were risk-based, they should be explained by these features.

2.3 AAA fraction and AAA adjustment

Let

$$AAA_i^{actual} = \text{fraction of CDO } i \text{ actually rated AAA},$$
$$AAA_i^{model} = \text{fraction of CDO } i \text{ implied by the CRA model}.$$

The paper defines the main adjustment as

$$Adj_i = AAA_i^{actual} - AAA_i^{model}. \tag{1}$$

Interpretation: A positive Adj_i means the final rating process allowed more AAA securities than the model output supported. This is the paper's central measure of subjectivity.

2.4 Criterion deviation

The paper also studies the difference between:

- the default probability criterion actually used in the surveillance report, and
- the publicized default probability criterion for the same rating and maturity.

A positive criterion deviation means the actual criterion is looser than the publicized criterion. The paper maps this into rating notches by asking whether the actual default probability would satisfy the publicized criterion for AAA, AA+, AA, and so on.

Interpretation: The adjustment variable is about how much AAA is added beyond the model. The criterion-deviation variable asks whether the benchmark used to judge AAA itself was looser than the public rule.

2.5 Future performance

The paper measures rating performance using:

- the number of notches downgraded from initial AAA by December 31, 2008;
- the time to downgrade to CC or below by June 30, 2010.

Interpretation: If adjustments reflected valuable analyst information, larger adjustments should be associated with fewer downgrades. The paper finds the opposite.

3 Models and empirical specifications

3.1 OLS determinants of AAA adjustments

The paper estimates regressions of the form

$$Adj_i = \alpha + \beta AAA_i^{model} + \gamma' X_i + \varepsilon_i, \tag{2}$$

where X_i includes collateral characteristics, CDO type dummies, closing-year dummies, manager experience, overcollateralization, insurance, liquidity support, multiple-CRA indicators, excess spread, and alternative model-implied AAA fractions.

Interpretation: The coefficient β tests whether subjective adjustments are systematically related to the CRA model's own predicted AAA fraction. A large negative β means the final rating process adds AAA precisely when the model predicts less AAA.

3.2 Downgrade regressions

For the downgrade test, the paper estimates ordered logit models:

$$DowngradeNotches_i^* = \alpha + \delta Adj_i + \theta' X_i + u_i, \tag{3}$$

with observed downgrade categories determined by threshold cutoffs.

Interpretation: A positive δ means CDOs with larger adjustments suffer more severe ex post downgrades. This is evidence against the view that adjustments incorporated useful soft information.

3.3 Hazard model for severe downgrades

The paper also estimates proportional hazard models for time to downgrade to CC or below:

$$h_i(t) = h_0(t) \exp\left(\delta Adj_i + \theta' X_i\right). \tag{4}$$

Interpretation: A positive adjustment coefficient means that larger adjustments are associated with faster collapse of initially AAA-rated CDO tranches into very low ratings.

3.4 Monte Carlo comparison model

The authors build a Monte Carlo model to compare the CRA output with a conventional Gaussian copula-style implementation based on collateral default probabilities, maturity, correlation, and asset counts.

Interpretation: The goal is not to prove the CRA model was technically wrong. It is to ask whether the CRA's own model output and a standard external implementation can explain the final AAA sizes. They cannot explain the adjustments.

3.5 Valuation calculation

The valuation effect is approximated by

$$ValueDiff = CollateralMaturity \times SizeAffected \times SpreadDifference. \tag{5}$$

Interpretation: The calculation asks how much the affected AAA portions would have been worth if priced at the lower model-implied rating rather than AAA. Even conservative methods imply large dollar effects.

4 Identification and empirical design

4.1 What identifies subjectivity

- The paper observes both final ratings and model-implied ratings.
- The adjustment is computed inside the rating process: actual AAA fraction minus CRA model AAA fraction.
- The authors then ask whether observable collateral features or alternative models explain this wedge.

Interpretation: The paper does not need to infer subjectivity from crisis losses alone. It directly measures deviations from the CRA model output.

4.2 Why cross-sectional variation matters

- Different CDOs have very different model-implied AAA fractions.
- If the final rating process were strictly model-based, actual AAA fractions would closely track model-implied AAA fractions.
- Instead, actual AAA fractions are much more compressed, and the adjustment is strongly negatively related to model-implied AAA.

Interpretation: The strongest pattern is not merely that AAA tranches are large. It is that low-model-AAA deals receive large positive adjustments.

4.3 Why downgrades matter

- A benign interpretation is that analysts used soft information to improve ratings beyond the model.
- If so, adjusted deals should perform at least as well as unadjusted deals.
- The paper finds that adjusted deals perform worse.

Interpretation: The ex post downgrade evidence is the key performance test of the adjustment mechanism.

4.4 Threats and limitations

- The data come from one major CRA rather than all rating agencies.
- The paper cannot observe all analyst discussions or internal motives.
- The valuation effects are suggestive and depend on spread benchmarks and assumptions about tranche priority.
- The Monte Carlo comparison may not exactly replicate every detail of the CRA model.

Interpretation: The paper's evidence is still unusually direct because it uses the CRA model's own output and links the adjustment to later performance.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent original figures and tables are grouped to save space and to make comparisons faster. Images are directly cropped from the paper.

5.1 Figure 1 and Table I: rating distribution and sample description

Read:

- Figure 1 Panel A shows that 84.08% of new CDO issuance is rated AAA, compared with 11.6% for corporate bonds.
- Figure 1 Panel B shows that by June 2010, only 29.1% of originally AAA CDOs remain AAA, while 45.2% have fallen to non-investment grade.
- Table I shows the sample is large: 916 CDOs, average collateral size $634.3 million, and average actual AAA fraction 75.5%.
- The sample is dominated by CLOs and ABS CDOs, with smaller numbers of CBOs and CDO^2.

Economic message: The crisis outcome is not a minor rating drift. CDOs were initially far more AAA-heavy than corporate bonds and later suffered much more severe rating deterioration.

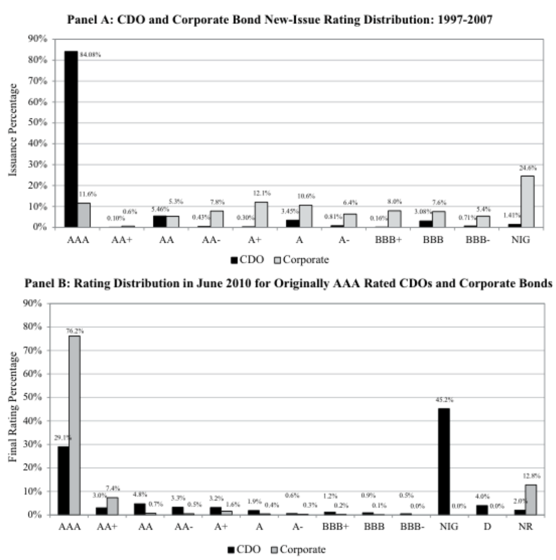


Figure 1. Credit rating distribution. The top figure (Panel A) plots the new-issue dollar value rating distribution for CDOs and corporate debentures issued between January 1997 and December 2007. The vertical axis is the issuance fraction in dollar value with the corresponding credit rating. “NIG” refers to noninvestment grade (ratings below BBB–). The corporate debentures consist

(a) Credit rating distribution.

Table I
CDO Sample Description

This table reports the average value of the collateral asset characteristics and liability structure for CDOs in our sample. Data are from the first CRA surveillance reports after closing. CDOs are issued over the January 1997 to December 2007 period. The last reporting date is September 2008. Data are grouped by collateral asset type (CBO for collateralized bond obligations, CLO for collateralized loan obligations, ABS CDO for CDOs of asset-backed securities, and CDO^2 for CDOs of CDOs). *Col. Rating* is the collateral asset average credit rating (average is calculated after numerical conversion AAA = 1, AA+ = 2, AA = 3, . . . C = 21). *Col. Default Rate* is the average expected collateral asset default rate in percentage. *Col. Maturity* is the collateral asset weighted average maturity. *Col. Size* is the total principal value of collateral assets. # *Assets* is the average number of assets in the collateral pool. # *Obligors* is the average number of distinctive obligors for the collateral assets. *Synthetic Dummy* equals one if the CDO is structured synthetically (using credit default swap, CDS, contracts), and zero if the CDO is a cash deal. *Mgr Deal* # is the average number of CDOs that the collateral manager has managed including the current CDO. *Overcollateralization* is the ratio of total collateral asset principal value over total liability principal value. *Insurance Dummy* equals one if the AAA tranche of the CDO is insured, and zero otherwise. *Liquidity Dummy* equals one if the CDO has liquidity facility (such as a revolving credit line or reserve account), and zero otherwise. *AAA Fraction* is the fraction of the CDO liability rated AAA; it counts super-senior tranches as AAA rated.

	CDO Type				
Variables	All	CBO	CLO	ABS CDO	CDO^2
# Obs.	916	96	393	373	54
Col. Rating	BB+	BB–	B+	A–	BBB
Col. Default Rate (%)	2.69	3.83	4.32	0.86	1.47
Col. Maturity (Years)	6.45	5.30	5.74	7.23	8.32
Col. Size ($ millions)	634.3	394.4	479.3	865.9	589.4
# Assets	218.3	139.2	325.9	144.7	84.02
# Obligors	130.0	104.3	158.1	115.3	72.1
Synthetic Dummy	0.14	0.25	0.00	0.25	0.15
Mgr Deal #	7.9	4.4	8.6	7.9	8.5
Overcollateralization	1.004	0.886	0.948	1.046	1.335
Insurance Dummy	0.061	0.188	0.043	0.048	0.056
Liquidity Dummy	0.235	0.469	0.112	0.284	0.370
AAA Fraction	0.755	0.728	0.726	0.798	0.715

(b) CDO sample description.

5.2 Table II and Figure 2: actual AAA versus model-implied AAA

Read:

- Table II reports that actual AAA fraction averages 75.5%, while the CRA model implies 63.4%.
- The average AAA adjustment is 12.1 percentage points.
- 770 of 916 adjustments are positive.
- Adjustments increase over time: 6.2% in 2003–2004, 9.7% in 2005, 12.8% in 2006, and 18.2% in 2007.

- The correlation between actual AAA and model AAA is only 0.49.
- The correlation between adjustment and model AAA is −0.71.
- Figure 2 shows that model-implied AAA fractions are centered around lower values, while actual AAA fractions bunch near 70–80%.

Economic message: The final rating outcome is not a tight translation of the model output. The adjustment systematically shifts many CDOs toward large and similar AAA tranche sizes.

Table II
CDO AAA Fraction: Actual, Credit Rating Agency Model, and Adjustment

This table reports the average value of the actual AAA fraction, the CRA model–predicted AAA fraction, and the adjustment (difference between actual and CRA model) for CDOs in our sample. Data are from first CRA CDO surveillance reports and the CDO rating databases. CDOs are issued over the January 1997 to December 2007 period. *Actual AAA* is the actual fraction of the CDO liability rated AAA, treating super-senior tranches as AAA. *CRA Model AAA* is the fraction of the CDO that can be rated AAA according to the rating agency model, defined as $1–SDR^{AAA}$. *CRA Adjustment* is the difference between the actual AAA fraction and CRA model AAA fraction. Panel A displays the sample average value. Data are grouped by collateral asset type (CBO for collateralized bond obligations, CLO for collateralized loan obligations, ABS CDO for CDOs of asset-backed securities, and CDO^2 for CDOs of CDOs) from the first CRA surveillance reports. Panel B displays the Pearson correlation matrix with *t*-statistics of the correlation coefficients in parentheses.

Panel A: Sample Average Value

Variables	All	CBO	CLO	ABS CDO	CDO^2
Obs.	916	96	393	373	54
Actual AAA	0.755	0.728	0.726	0.798	0.715
CRA Model AAA	0.634	0.625	0.566	0.717	0.568
CRA Adjustment	0.121	0.104	0.160	0.081	0.147
Positive/Total	770/916	75/96	384/393	263/373	48/54
CRA Adjustment ≤ 2002	0.107	0.106	0.127	0.066	0.127
Positive/Total	102/131	52/65	21/25	24/36	21/25
CRA Adjustment 2003 to 2004	0.062	0.064	0.129	0.003	0.129
Positive/Total	118/155	3/4	67/69	42/74	67/69
CRA Adjustment 2005	0.097	–0.035	0.149	0.057	0.019
Positive/Total	136/156	1/2	73/73	55/73	7/8
CRA Adjustment 2006	0.128	0.091	0.154	0.101	0.128
Positive/Total	223/261	8/11	126/127	79/110	10/13
CRA Adjustment 2007	0.182	0.133	0.206	0.153	0.219
Positive/Total	194/213	11/14	98/99	65/80	20/20

Panel B: Pearson Correlations

Variables	Actual AAA	CRA Model AAA
CRA Model AAA	0.49(14.09)	
CRA Adjustment	0.27(5.34)	−0.71(−16.34)

To examine the effect of the adjustment on the overall AAA graphically

(a) Actual, model-implied, and adjusted AAA fractions.

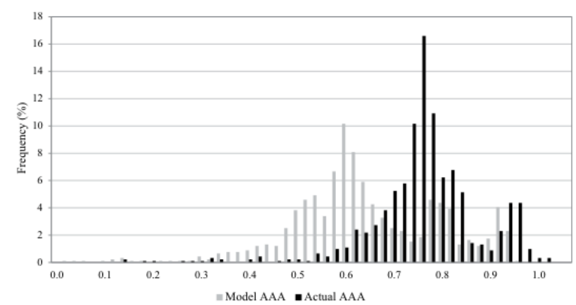


Figure 2. Distribution of AAA fraction from credit rating agency model and actual AAA fraction in the capital structure of the CDO. This figure reports the histograms of the AAA fraction from the output of the CRA model (gray bars) and the fraction of the CDO actually rated AAA (black bars). The sample includes 916 CDOs issued between January 1997 and December 2007. The test for differences in the distribution of AAA fraction across two groups is conducted by calculating the corrected Kolmogorov–Smirnov *D*-statistic, with *p*-values of the test less than 0.0001. The simple mean difference test between model AAA and actual AAA has a *t*-statistic of 25.93.

(b) Distribution of model and actual AAA fractions.

5.3 Figures 3 and 4: the adjustment is largest when the model gives little AAA

Read:

- Figure 3 shows a strong negative relation between model-implied AAA fraction and the initial adjustment.
- Cash and synthetic CDOs both show the same broad pattern.
- Figure 4 sorts CDOs into quintiles by model-implied AAA fraction.
- Deals in the lowest model-AAA quintile receive large additions to AAA, while high model-AAA deals receive little adjustment.
- The paper reports that the lowest quintile has 42.6% model AAA plus a 26.8% adjustment, producing 69.4% actual AAA.
- For CDO^2, the lowest quintile has 29.2% model AAA plus a 47.0% adjustment, producing 76.1% actual AAA.

Economic message: The adjustment acts like a compression device. It moves weak model deals closer to the typical AAA structure instead of letting model-implied differences fully appear in final ratings.

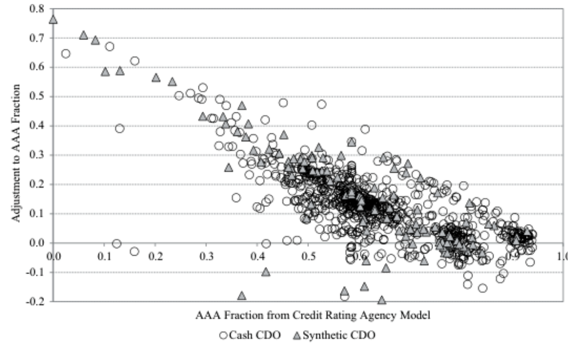


Figure 3. Credit rating agency model–predicted AAA fraction (*x*-axis) and initial adjustment (*y*-axis). This figure graphs the AAA fraction from the CRA model (defined as 1–SDR^{AAA}) in the first surveillance reports and the adjustment (difference between actual CDO fraction–rated AAA and credit rating agency model AAA fraction). SDR^{AAA} is the scenario default rate for the AAA scenario directly from the rating agency model output. The sample includes 916 CDOs issued

(a) Model-predicted AAA and adjustment.

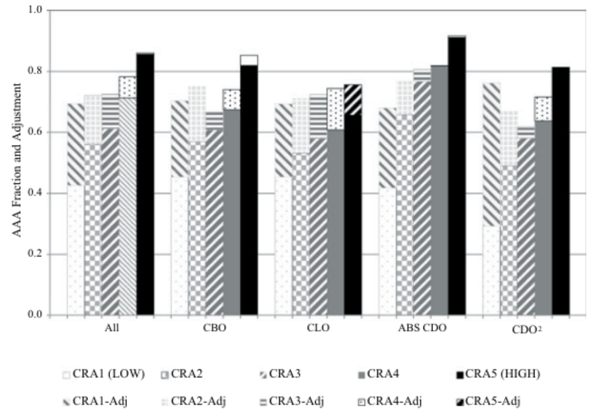


Figure 4. Credit rating agency model–predicted AAA fraction and initial adjustment by collateral asset type. This figure graphs the AAA fraction from the CRA model (defined as 1–SDR^{AAA}) in the first surveillance reports and the adjustment (difference between actual CDO fraction–rated AAA and CRA model AAA fraction). The bottom bars are from the CRA model's AAA fractions, and the top bars are adjustments. The total length of the bars is the actual AAA fraction. Data are divided into different collateral asset types (CBO, CLO, ABS CDO, CDO^2). Within each CDO type, the data are further separated into five groups according to CRA model AAA fraction.

(b) AAA fraction and adjustment by collateral type.

5.4 Table III: do observable characteristics explain adjustments?

Read:

- Manager experience alone has tiny explanatory power.
- Overcollateralization has a negative sign in many specifications.
- Insurance, liquidity support, Vasicek AAA, simulation AAA, multiple-CRA status, and excess spread do not explain away the adjustment.
- The CRA model AAA fraction is the dominant predictor: the coefficient is strongly negative and statistically significant.
- Adding the model AAA fraction raises adjusted R^2 substantially.

Economic message: The most important determinant of the adjustment is the model's own low AAA output. This is difficult to reconcile with adjustments being purely risk-based corrections for missing deal quality.

5.5 Tables IV and V: adjustments predict worse subsequent downgrades

Read:

- Table IV uses ordered logit regressions for the number of downgrade notches by December 31, 2008.
- AAA adjustment is positive and statistically significant across specifications.
- The result survives controls for CDO type, vintage/type interactions, overcollateralization, insurance, liquidity, multiple CRAs, and shock dates.
- Table V uses a hazard model for downgrade to CC or below by June 30, 2010.
- The adjustment again predicts faster and more severe downgrading.

Economic message: The performance evidence rejects the benign interpretation that adjustments incorporated superior qualitative information. Larger adjustments are associated with worse future rating outcomes.

Table III

CRA AAA Fraction Adjustment and CDO Characteristics

This table shows the results of OLS regressions. The dependent variable is the CRA AAA fraction adjustment. The adjustment is defined as the difference between the actual AAA fraction and the CRA model–predicted AAA fraction explained in Table II. The independent variables are described in Table I except the following: *Vasicek AAA* is the AAA fraction of the CDO predicted by the Vasicek model, *Simulation AAA* is the AAA fraction predicted by a Monte Carlo simulation, *Multiple CRA* is a dummy variable with one for multiple rating on the CDO, and zero otherwise, *Excess Spread* is the ratio of the average collateral coupon rate over the average CDO note coupon rate, and *BDR – SDR* is the difference between the break-even default rate (BDR) and scenario default rate (SDR) in the presale or new-issue reports. Data are from CRA CDO presale, new-issue, and surveillance reports, as well as CDO rating databases. CDOs are issued over the January 1997 to December 2007 period. White (1980) heteroskedasticity-adjusted t-statistics are in parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.107	0.182	0.529	0.559	0.574	0.531	0.498	0.654	0.414
	(13.23)	(16.14)	(38.03)	(38.46)	(41.93)	(24.90)	(22.81)	(17.68)	(10.61)
Log(Mgr Deal #)	0.010	0.012		0.007		0.005	0.003	0.001	0.005
	(2.21)	(2.82)		(2.44)		(1.60)	(0.87)	(0.34)	(1.23)
Overcollateralization		−0.079		−0.061	−0.061	−0.063	−0.060	−0.097	−0.030
		(−10.07)		(−11.00)	(−11.03)	(−11.20)	(−10.89)	(−13.21)	(−5.48)
Insurance Dummy		0.034		0.040		0.045	0.049	0.055	0.035
		(1.83)		(3.10)		(3.47)	(3.79)	(3.99)	(1.69)
Liquidity Dummy		−0.005		0.006		0.008	0.014	0.013	−0.002
		(−0.43)		(0.75)		(1.06)	(1.84)	(1.77)	(−0.21)
CRA AAA			−0.642	−0.618	−0.618	−0.737	−0.721	−0.778	−0.596
			(−30.18)	(−30.83)	(−30.74)	(−19.57)	(−19.31)	(−19.58)	(−9.91)
Vasicek AAA						0.030	−0.022	−0.017	0.054
						(1.46)	(−1.01)	(−0.73)	(1.48)
Simulation AAA						0.112	0.139	0.042	0.032
						(2.55)	(3.17)	(0.91)	(0.53)
CLO						0.035	0.025	0.029	0.073
						(2.81)	(1.97)	(1.95)	(2.31)
ABS CDO						0.031	0.054	0.085	0.032
						(1.81)	(3.09)	(4.28)	(0.83)
CDO^2						0.017	0.033	0.012	0.005
						(0.79)	(1.55)	(0.53)	(0.13)
Synthetic Dummy						−0.009	−0.025	−0.035	−0.028
						(−0.81)	(−2.20)	(−2.95)	(−1.09)
Closing Year 2005							0.020	0.011	0.027
							(2.04)	(1.13)	(2.51)
Closing Year 2006							0.029	0.011	0.045
							(3.15)	(1.25)	(4.28)
Closing Year 2007							0.059	0.035	0.069
							(5.56)	(3.28)	(4.97)
Multiple CRAs								−0.001	0.021
								(−0.08)	(1.70)
Excess Spread								−0.013	
								(−2.51)	
BDR – SDR									−0.004
									(−3.63)
N	903	903	903	903	903	903	903	669	408
Adjusted R^2	0.005	0.112	0.503	0.569	0.562	0.584	0.598	0.696	0.638

Table IV

AAA Fraction Adjustment and Subsequent Downgrading as of December 31, 2008

This table shows ordered logit regression results. The dependent variable is the number of notches downgraded from initial AAA rating. *AAA Adjustment*, the first independent variable, is defined as the difference between the actual AAA fraction with super-senior tranches and the CRA model–predicted AAA fraction as described in Table II, truncated on the left side at zero. This variable is collected from the first surveillance report data after issuance. *Multiple CRA* is a dummy variable with one for multiple ratings on the CDO, and zero otherwise. *ABS 2004–2007* and *CDO*^{2} are interactions between CDO type and closing year. *Shocks* represents the five dates when the most CDOs are downgraded. For example, the first three occur when CDOs backed by various types of RMBS collateral are placed on credit watch or downgraded. Other independent variables are described in Table I. Data are from the first CRA CDO surveillance reports and the CDO rating databases. CDOs are issued over the January 1997 to December 2007 period. Odds ratios are reported with White (1980) heteroskedasticity-adjusted z-statistics in parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)
AAA Adjustment	68.19	32.14	15.60	17.38	18.21	9.46
	(5.30)	(4.18)	(3.35)	(3.46)	(3.49)	(3.08)
ABS CDO	80.69	74.76	27.41	26.12	25.70	28.88
	(14.50)	(14.51)	(9.25)	(9.16)	(9.12)	(8.97)
CDO^2	76.19	73.32	3.45	1.91	1.85	1.80
	(10.08)	(9.72)	(1.01)	(0.78)	(0.74)	(0.75)
Synthetic Dummy		3.21	1.47	1.16	1.17	1.42
		(4.45)	(1.22)	(0.46)	(0.47)	(1.19)
ABS 2004			1.21	1.07	1.10	1.14
			(0.52)	(0.19)	(0.26)	(0.32)
ABS 2005			4.11	4.70	4.77	5.32
			(4.58)	(4.95)	(5.02)	(4.92)
ABS 2006			12.16	12.51	12.44	23.43
			(6.59)	(6.69)	(6.68)	(8.64)
ABS 2007			29.96	35.75	36.22	62.68
			(7.96)	(8.15)	(8.21)	(9.97)
CDO^2 2004			7.35	4.28	4.29	5.70
			(1.29)	(1.03)	(1.03)	(1.38)
CDO^2 2005			22.01	36.64	37.00	40.83
			(2.21)	(3.27)	(3.27)	(3.48)
CDO^2 2006			41.47	77.43	77.70	108.56
			(2.68)	(4.26)	(4.25)	(4.54)
CDO^2 2007			327.73	619.05	655.52	1130.70
			(4.08)	(5.92)	(5.93)	(7.01)
Overcollateralization				1.22	1.21	1.25
				(1.81)	(1.78)	(2.29)
Insurance Dummy				1.22	1.25	1.55
				(0.51)	(0.55)	(0.99)
Liquidity Dummy				2.00	2.01	1.78
				(3.19)	(3.18)	(2.71)
Multiple CRAs					1.57	1.30
					(0.97)	(0.56)
Shocks	Y	Y	Y	Y	Y	N
N	916	916	916	916	916	916
Pseudo R^2	0.313	0.322	0.368	0.373	0.374	0.359

(a) Ordered logit model of subsequent downgrading.

Table V

Hazard Model of AAA Downgrading as of June 30, 2010

This table shows estimation results for the downgrading hazard model. The dependent variable is time to a downgrade of CC or below for initially AAA-rated CDOs. *AAA Adjustment*, the first independent variable, is defined as the difference between the actual AAA fraction with super-senior tranches and the CRA model-predicted AAA fraction as described in Table II, truncated on the left side at zero. *Multiple CRA* is a dummy variable with one for multiple ratings on the CDO, and zero otherwise. Other independent variables are described in Table IV. Data are from the first CRA CDO surveillance reports and the CDO rating databases. CDOs are issued over January 1997 to December 2007 period. Hazard ratios are reported with White (1980) heterosketasticity-adjusted t-statistics in parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)
AAA Adjustment	6.13	2.93	2.62	2.83	2.82	3.86
	(4.29)	(2.43)	(2.38)	(2.45)	(2.44)	(3.22)
ABS CDO	5.33	5.48	1.22	1.23	1.22	3.07
	(5.98)	(6.54)	(0.58)	(0.55)	(0.53)	(3.27)
CDO^2	8.32	9.29	0.63	0.50	0.49	0.82
	(6.63)	(7.33)	(–0.46)	(–0.67)	(–0.68)	(–0.19)
Synthetic Dummy		3.00	1.22	1.11	1.11	1.20
		(6.58)	(1.05)	(0.57)	(0.56)	(0.94)
ABS 2004			4.03	3.83	3.85	4.28
			(4.65)	(4.42)	(4.43)	(4.38)
ABS 2005			12.52	13.49	13.52	15.12
			(8.71)	(8.75)	(8.77)	(8.50)
ABS 2006			45.68	46.54	46.52	68.93
			(11.56)	(11.49)	(11.49)	(12.38)
ABS 2007			211.89	230.82	233.11	322.94
			(13.59)	(13.89)	(13.91)	(14.52)
CDO^2 2004			8.54	6.48	6.50	9.71
			(2.01)	(1.75)	(1.76)	(2.12)
CDO^2 2005			42.99	52.01	51.94	79.20
			(3.77)	(3.78)	(3.78)	(4.23)
CDO^2 2006			84.48	118.99	118.72	158.44
			(4.24)	(4.42)	(4.42)	(4.63)
CDO^2 2007			427.31	578.98	581.08	744.70
			(5.76)	(5.76)	(5.76)	(5.90)
Overcollateralization				1.13	1.13	1.16
				(2.12)	(2.11)	(2.66)
Insurance Dummy				1.06	1.06	1.32
				(0.15)	(0.16)	(0.76)
Liquidity Dummy				1.42	1.41	1.36
				(2.60)	(2.56)	(2.28)
Multiple CRAs					1.16	1.22
					(0.44)	(0.51)
Shocks	Y	Y	Y	Y	Y	N
N	916	916	916	916	916	916

and Shumway (2008) and use a proportional hazard model to examine the relation between adjustments and the likelihood of an AAA security downgrade

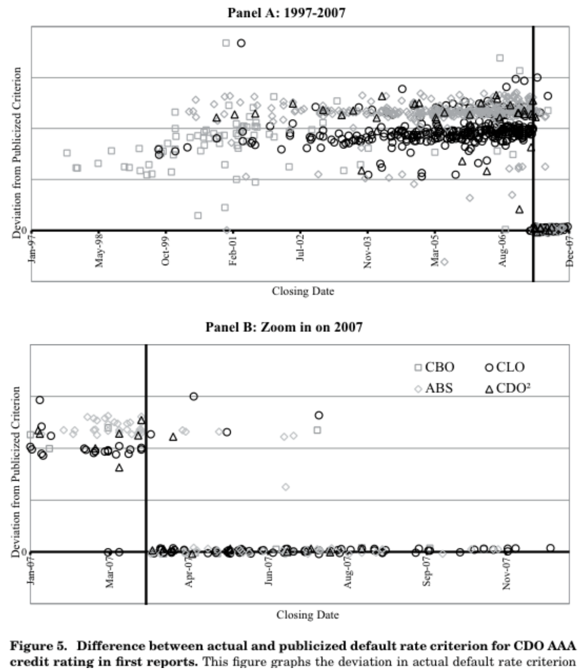
(b) Hazard model of downgrade to CC or below.

5.6 Figures 5 and 6: default-rate criterion deviation

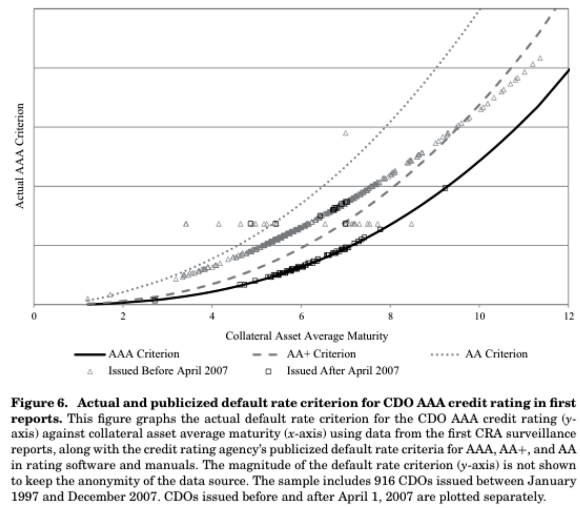
Read:

- Figure 5 plots the difference between the actual and publicized default-rate criterion for AAA CDO ratings.
- Before April 2007, many observations show positive deviations, meaning the actual criterion was looser than the publicized criterion.
- After April 2007, most first-report observations move close to zero deviation.
- Figure 6 plots actual default-rate criteria against maturity and compares them with publicized AAA, AA+, and AA curves.
- Before April 2007, most actual AAA criteria lie closer to the publicized AA standard than to the publicized AAA standard.

Economic message: The issue is not only subjective adjustment around a model. The paper also finds evidence that the default-rate criterion used to grant AAA was itself looser before April 2007.



(a) Deviation from publicized AAA default-rate criterion.



(b) Actual and publicized criteria in first reports.

5.7 Table VI and Figure 7: mapping criteria into rating notches

Read:

- Table VI maps actual default criteria into publicized rating notches.
- Before April 1, 2007, only 1.3% of AAA-rated CDOs comply with the publicized AAA criterion.
- 4.8% comply with AA+ and 92.5% comply with AA.
- After April 1, 2007, compliance rates exceed 90% across ratings.
- Figure 7 shows that continuing surveillance reports after April 2007 still have two patterns: new post-April-2007 deals tend to follow the publicized criteria, while many pre-April deals retain the old criterion.

Economic message: The default-criterion evidence quantifies rating inflation in rating-notch terms. Before April 2007, many AAA CDO ratings looked like AA ratings under the publicized criterion.

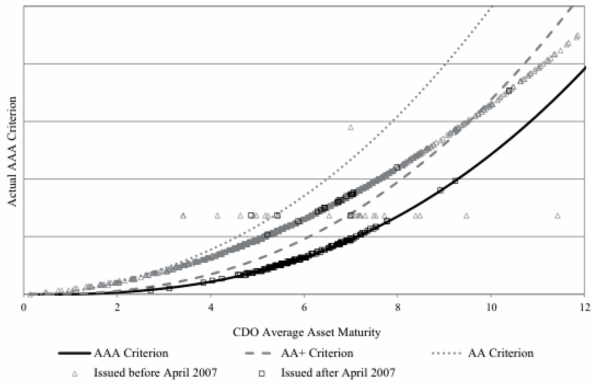


Figure 7. Actual and publicized default rate criterion for CDO AAA credit rating in all reports after April 1, 2007. This figure graphs the actual default rate criterion for the CDO AAA credit rating (*y*-axis) against collateral asset average maturity (*x*-axis) using data from all CRA surveillance reports from April 1, 2007 to September 30, 2008, along with the CRA's publicized default rate criteria for AAA, AA+, and AA in rating software and manuals. The magnitude of the default rate criterion (*y*-axis) is not shown to keep the anonymity of the data source. The sample includes 916 CDOs issued between January 1997 and December 2007. CDOs issued before and after April 1, 2007 are plotted separately.

Actual-Publicized CDO Rating Criterion Mapping Matrix

table reports the frequency distribution of rating default rate criterion deviations. In the row "AAA," we compare the AAA rating default rat rion actually reported in rating agency surveillance reports to the publicized default rate criterion with the same maturity in publicly distribute rating software and manuals. If the actual criterion qualifies the publicized criterion, then we assign "0" notches deviated. If the actual criterio not qualify for AAA rating according to publicized criterion, but qualifies for the "AA+" ("AA", "AA–," "A+" and below) criterion, then we assig ("–2," "–3," "–4 or less") notches deviated. We calculate the percentage for each category of notch deviations, so all numbers in each row add up t This procedure is done for all ratings from "AAA" to "CCC–." A default probability can qualify for a certain rating if the actual default probabilit ss than 1.05 times the publicized default probability of such a rating. The default probabilities are rounded to four decimal points. CDOs ar d over the periods from January 1997 to March 2007 (Panel A) and from April 2007 to December 2007 (Panel B).

Panel A: CDOs Issued before April 1, 2007								
	Number of Notches Deviated							
	3 or more	2	1	0	−1	−2	−3	−4 or les
	–	–	–	0.013	0.048	0.925	0.009	0.005
	–	–	0.005	0.017	0.951	0.015	0.008	0.004
	–	0.001	0.001	0.022	0.018	0.135	0.717	0.106
	0.000	0.001	0.017	0.010	0.041	0.447	0.461	0.023
	0.001	0.001	0.008	0.015	0.066	0.860	0.046	0.003
	0.001	0.003	0.012	0.021	0.735	0.225	0.004	0.000
	0.001	0.004	0.014	0.049	0.923	0.006	0.003	0.000
+	0.004	0.009	0.008	0.408	0.568	0.004	0.000	0.000
	0.004	0.009	0.009	0.965	0.013	0.000	0.000	0.000
−	0.012	0.001	0.008	0.976	0.004	0.000	0.000	0.000
	0.012	0.001	0.015	0.959	0.010	0.003	0.000	0.000
	0.012	0.003	0.006	0.964	0.013	0.003	0.000	0.000
	0.013	0.001	0.073	0.909	0.004	0.000	0.000	0.000
	0.013	0.006	0.906	0.071	0.004	0.000	0.000	0.000
	0.014	0.012	0.499	0.467	0.008	0.000	0.000	0.000
	0.019	0.031	0.866	0.080	0.004	0.000	0.000	–
+	0.022	0.059	0.875	0.044	0.000	0.000	–	–
	0.022	0.492	0.483	0.004	0.000	–	–	–
−	0.013	0.637	0.346	0.004	–	–	–	–

(continued

(a) Table VI Panel A: before April 1, 2007.

Table VI—*Continued*

Panel B: CDOs Issued after April 1, 2007								
	Number of Notches Changed							
	3 or more	2	1	0	−1	−2	−3	−4 or les
	–	–	–	0.913	0.008	0.072	0.007	0.000
	–	–	0.000	0.913	0.072	0.007	0.007	0.000
	–	0.000	0.000	0.913	0.000	0.007	0.065	0.014
	0.000	0.000	0.000	0.913	0.007	0.058	0.014	0.007
	0.000	0.000	0.000	0.913	0.007	0.065	0.014	0.000
	0.000	0.000	0.000	0.913	0.072	0.014	0.000	0.000
	0.000	0.000	0.000	0.920	0.072	0.007	0.000	0.000
+	0.000	0.000	0.000	0.957	0.043	0.000	0.000	0.000
	0.000	0.000	0.000	0.986	0.014	0.000	0.000	0.000
−	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.986	0.014	0.000	0.000	0.000
	0.000	0.000	0.000	0.986	0.014	0.000	0.000	0.000
	0.000	0.000	0.007	0.993	0.000	0.000	0.000	0.000
	0.000	0.000	0.072	0.928	0.000	0.000	0.000	0.000
	0.000	0.000	0.072	0.928	0.000	0.000	0.000	0.000
	0.000	0.000	0.072	0.928	0.000	0.000	0.000	–
+	0.000	0.000	0.087	0.913	0.000	0.000	–	–
	0.000	0.065	0.022	0.913	0.000	–	–	–
−	0.000	0.072	0.014	0.913	–	–	–	–

(b) Table VI Panel B: after April 1, 2007.

5.8 Table VII and Table VIII: simulation comparison and valuation effects

Read:

- Table VII compares the CRA model AAA fraction with the authors' Monte Carlo simulation.
- Collateral characteristics explain much of the difference, but the comparison does not eliminate the main adjustment finding.
- Table VIII translates adjustments and criterion deviations into valuation effects.
- Method 1 implies a value difference of $14.73 million per CDO, or $12.33 million excluding criterion deviation.
- Method 2 implies $42.20 million per CDO, or $35.02 million excluding criterion deviation.
- Bundling AAA tranches together implies $94.13 million per CDO, or $78.91 million excluding criterion deviation.

Economic message: Even the conservative valuation method implies that subjectivity was economically large relative to underwriting fees. The more investor-oriented methods imply much larger dollar magnitudes.

Table VII
Rating Agency and Our Simulation AAA Fraction Difference Regressions

This table reports coefficient estimates from OLS regressions. The dependent variables are the difference between the CDO AAA fraction from the CRA model and from a Monte Carlo model (CRA AAA – Monte Carlo AAA). The Monte Carlo simulation inputs are average collateral default rates, maturity, correlations, and number of assets reported by the CRA. The simulation approach is described in the Internet Appendix. The independent variables are described in Table I. *CLO*, *ABS CDO*, and *CDO²* are collateral asset type dummy variables. Closing year dummies for 2002 to 2007 are included with closing year 2001 and before as the comparison group. CDOs are issued over the January 1997 to December 2007 period. White (1980) heteroskedasticity-adjusted *t*-statistics are in parentheses.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Intercept	0.160	0.016	0.036	−0.006	0.137	0.471	0.562	0.201
	(1.68)	(3.17)	(5.83)	(−0.63)	(1.37)	(4.24)	(6.07)	(1.94)
Col. Def. Prob.	−2.116				−2.118		−1.400	−2.168
	(−13.05)				(−12.99)		(−11.09)	(−13.25)
Avg. Col. Rating	0.008				0.008	−0.002		0.009
	(7.28)				(6.93)	(−2.00)		(7.11)
Avg. Col. Maturity	−0.007				−0.007	−0.003	−0.006	−0.006
	(−4.16)				(−4.16)	(−1.80)	(−3.52)	(−3.68)
Correlation	−0.003				−0.005	−0.026	−0.019	−0.036
	(−0.18)				(−0.27)	(−1.18)	(−0.93)	(−1.77)
Log(CDO Size)	−0.011				−0.011	−0.027	−0.029	−0.015
	(−2.44)				(−2.35)	(−4.94)	(−6.11)	(−3.00)
Number of Assets	0.000				0.000	0.000	0.000	−0.000
	(1.21)				(0.11)	(0.63)	(0.44)	(−0.05)
Number of Obligors	0.001				0.001	0.001	0.001	0.001
	(10.07)				(10.32)	(9.76)	(10.10)	(10.79)
Log(Mgr Deal #)		0.009			−0.003	−0.004	−0.004	−0.004
		(3.13)			(−1.42)	(−1.69)	(−1.52)	(−1.65)
Overcollateralization			−0.002		0.003	0.005	0.003	0.004
			(−0.30)		(0.81)	(1.19)	(0.81)	(1.11)
Insurance Dummy			−0.033		0.003	−0.011	0.002	0.004
			(−2.69)		(0.27)	(−1.05)	(0.21)	(0.45)
Liquidity Dummy			−0.016		0.002	0.006	0.004	0.005
			(−2.31)		(0.40)	(0.97)	(0.70)	(0.90)
CLO				0.062	0.030	0.030	0.044	0.028
				(6.32)	(3.25)	(2.51)	(4.00)	(2.54)
ABS CDO				0.028	0.031	0.038	0.020	0.038
				(2.85)	(2.86)	(2.95)	(1.65)	(3.20)
CDO²				−0.027	0.014	0.019	0.005	0.016
				(−1.88)	(0.99)	(1.15)	(0.32)	(1.08)
Synthetic Dummy				−0.016	−0.007	0.001	0.002	−0.010
				(−1.85)	(−0.90)	(0.10)	(0.27)	(−1.23)
Closing Year 2002						0.003	−0.013	−0.009
						(0.17)	(−0.91)	(−0.68)
Closing Year 2003						0.020	0.006	0.014
						(1.47)	(0.43)	(1.14)
Closing Year 2004						0.002	−0.007	0.005
						(0.14)	(−0.62)	(0.46)
Closing Year 2005						0.009	−0.001	0.010
						(0.73)	(−0.05)	(0.90)
Closing Year 2006						−0.004	−0.010	0.002
						(−0.36)	(−0.91)	(0.19)
Closing Year 2007						0.019	0.017	0.029
						(1.44)	(1.37)	(2.46)
N	902	902	902	902	902	902	902	902
Adjusted R^2	0.452	0.010	0.011	0.099	0.460	0.364	0.439	0.469

(continued)

Table VII—*Continued*

(a) CRA versus Monte Carlo AAA fraction difference.

Table VIII
Valuation Effect of Subjectivity

This table reports the valuation consequence of adjustment and criterion deviation. The *Value Difference* per CDO is calculated as follows: *Value Difference* = *Collateral Average Maturity* × *Size Affected* × *Spread Difference*, where *Collateral Average Maturity* is the average maturity of collateral assets, *Size Affected* is the dollar value affected by subjectivity, and *Spread Difference* is the difference between the AAA spread of the affected part and a benchmark spread. The benchmark spread is calculated using three methods. The first method in Panel A takes CRA outputs across the entire rating spectrum and assigns ratings to the actual CDO structure. If the top tranche size is 75.5% and the rating agency model indicates 75.0% can obtain A rating, then this tranche is rated as BBB+. The second method in Panel B and its variant in Panel C use our Monte Carlo simulation to rate the tranches. Panel B evaluates each tranche separately. Panel C bundles all AAA tranches together. Two effects drive the difference between Panels B and C. First, on average, the junior AAA tranche has wider spread (46 bps) than the senior AAA tranche (35 bps). Second, junior AAA tranche size is on average smaller than the senior AAA tranche size. All methods are applied with and without the criterion deviation. Data are from the first CRA CDO surveillance reports and the CDO rating databases. Data are grouped by collateral asset type (*CBO* for collateralized bond obligations, *CLO* for collateralized loan obligations, *ABS CDO* for CDOs of asset-backed securities, and *CDO²* for CDOs of CDOs). The sample includes 916 CDOs issued over the January 1997 to December 2007 period.

	All	CBO	CLO	ABS CDO	CDO²
Number of CDOs	916	96	393	373	54
Collateral Average Maturity (years)	6.45	5.30	5.74	7.23	8.31
Panel A: Method #1 (Adjusted Portion with Model Implied Rating)					
Size Affected ($ millions)	87.67	56.51	83.90	98.37	96.52
Model Rating	BBB+	BBB	BBB−	A−	A
Spread Difference Total (%)	2.24	2.27	2.52	2.04	1.50
Value Difference Total ($ millions)	14.73	7.38	13.12	17.72	15.25
Value Difference w/o Deviation ($ millions)	12.33	6.12	11.96	14.08	11.17
Panel B: Method #2 (Multiple Tranches Impacted Differently)					
Size Affected ($ millions)	289.57	223.23	273.64	328.59	253.98
Model Rating	BBB	BBB	BBB	BBB	BBB+
Spread Difference Total (%)	2.40	2.51	2.49	2.41	1.57
Value Difference Total ($ millions)	42.20	29.46	39.35	48.74	32.77
Value Difference w/o Deviation ($ millions)	35.02	24.50	35.53	38.14	22.77
Panel C: Variant of Method #2 (Bundling AAA Together)					
Size Affected ($ millions)	529.62	342.51	366.54	761.05	450.60
Model Rating	BBB	BBB	BBB	BBB	BBB+
Spread Difference Total (%)	2.56	2.54	2.53	2.66	2.14
Value Difference Total ($ millions)	94.13	54.75	52.94	146.83	74.68
Value Difference w/o Deviation ($ millions)	78.91	47.92	47.49	120.40	57.47

(b) Valuation effect of subjectivity.

6 Short summary

- The paper studies whether subjectivity affected CDO ratings before the financial crisis.
- It uses direct evidence from a major CRA's model outputs and final ratings for 916 CDOs.
- The central measure is the AAA adjustment: actual AAA fraction minus model-implied AAA fraction.
- Actual AAA fraction averages 75.5%, while model-implied AAA averages 63.4%.
- The average adjustment is 12.1 percentage points, and 84.6% of adjustments are positive.
- Adjustments rise over time and reach 18.2% in 2007.
- Adjustments are largest when the model-implied AAA fraction is low.
- Standard deal features and alternative model outputs do not explain the adjustment.
- Larger adjustments predict worse subsequent downgrades.
- Before April 2007, the actual default-rate criterion used for AAA often corresponded more closely to the publicized AA criterion.
- The economic value of adjustments and criterion deviations is large, ranging from tens to almost one hundred million dollars per CDO depending on method.
- The main takeaway is that CDO ratings were not purely mechanical model outputs. Subjective adjustments and criterion deviations increased AAA issuance and were associated with poor ex post performance.

7 Conclusion

7.1 Core conclusion

Griffin and Tang provide direct evidence that subjectivity played an important role in CDO credit ratings. The central fact is that final AAA tranche sizes were systematically larger than the CRA's own main model implied. These positive adjustments were common, increasingly large over time, difficult to explain with standard deal characteristics, and associated with worse future downgrades.

7.2 Economic intuition

The economics are straightforward. A CDO is easier to sell and more profitable to structure if a larger portion of the capital structure receives a AAA rating. When the model implies a small AAA share, subjective adjustment can increase the AAA share and reduce the amount of lower-rated debt that must be placed with investors. This is especially important because lower tranches are harder to sell and require much higher spreads.

7.3 Takeaway

The paper's lesson is that credit-rating failures were not only about bad house-price forecasts or rare macroeconomic shocks. The rating process itself had measurable discretionary components. Those components increased AAA issuance and predicted worse performance, making them central to understanding structured-finance rating inflation before the crisis.